

8 Q1) choose the best answer

1. The output of a system $y(t)$ is related to its input $x(t)$ according to the relation $y(t) = x(t)\sin(2\pi t)$ this system is

- a- Linear and time-invariant
- b- Linear and time-variant
- c- Nonlinear and time-invariant
- ☒ d- Nonlinear and time-variant

2. Consider the system represented by $y[n] = 2x[n] + 3$

- a. The system is linear and follows the additive property.
- ☒ b. The system is linear and does not have homogeneity property.
- c. The system is nonlinear and does not have homogeneity property.
- d. The system does not have the homogeneity property but follows the additive property.

3. Consider a signal $x(t) = 5 \cos\left(\frac{2\pi t}{3}\right) + 9 \sin(0.5\pi t) + 3 \sin\left(\frac{\pi t}{3} - \frac{\pi}{6}\right) - 12$, so that

- a. $x(t)$ is periodic with frequency $(1/12)$ Hz.
- b. $x(t)$ is periodic with period 1.2s
- ☒ c. $x(t)$ is not periodic
- d. $x(t)$ is periodic with frequency 12 Hz

4. What's the even part of the signal $x(t) = 2 + \cos(t)$?

- a. $2 + \sin(t)$
- ☒ b. $2 + \cos(t)$
- c. $2 - \sin(t)$
- d. $2 \cos(t)$

5. Which of the following is false for a continuous time signal?

- a. Any signal can be expressed as the sum of an odd and an even signals.
- ☒ b. Sum of two periodic signals is always periodic.
- c. Any periodic signal can be expressed as sum of sinusoidal signals.
- d. None of the above.

6. If $f(t)$ has an energy E so the energy of $f(2t)$ and $f(0.5t)$ respectively are

- ☒ a. E and E
- b. E and $0.5E$
- c. $0.5E$ and $2E$
- d. $2E$ and $0.5E$

$-15) - r(t-15)$

7. The following systems $y(t) = \frac{1}{3}(x[n] + x[n-1] + x[n+2])$ is ☒ a. dynamic & non-causal
 a. Dynamic & causal b. static & non-causal c. static & causal

8. The value of $\int_{-\infty}^{\infty} e^{-t} \delta(2t-2) dt$ is ☒ a. $\frac{1}{2e}$ b. $\frac{2}{e}$ c. $\frac{1}{e^2}$ d. $\frac{1}{2e^2}$

9. If $x[n] = \{3, 2, 1\}$ and $h[n] = \left(\frac{1}{2}\right)^n u[n]$ if $y[n] = x[n] * h[n]$ so $y[1] = \frac{3}{2} + 2$ ☒ a. 0 b. 3 c. $\frac{3}{2}$ d. none of above

10. If $g(t) = \begin{cases} \frac{t+4}{4}, & -4 \leq t < 0 \\ \frac{2-t}{2}, & 0 \leq t < 2 \end{cases}$, so the $g(0.25t)$ will define in negative time period is ☒ a. $-1 \leq t < 0$ b. $-4 \leq t < 0$ c. $-16 \leq t < 0$ d. all are correct

11. for $\int_{-5}^6 x(t) \delta'(t-3) dt$ the result will be $= x(t) = t^2 + 2$ ☒ a. 0 b. -8 c. 11 d. 1

12. The $\text{sgn}(t) =$ ☒ a. $u(t) - u(t)$ b. $u(t) + u(-t)$ c. $u(t) - u(-t)$ d. $\text{sgn}(-t)$
 Answer the questions (13-15) through the convolution between $x(t) = 2$ only for $0 \leq t < 1$
 and $h(t) = 1$ only for $0 \leq t < 2$ where $y(t) = x(t) * h(t)$

13. $y(2) =$ ☒ a. 0 b. 2 c. 1 d. 0.5
 14. $y(t) = x(0.5) * h(0.5)$ equals to ☒ a. 0 b. 2 c. 1 d. 0.5
 15. The no overlapping area will be ☒ a. $t < 0$ only b. $t > 2$ only c. $0 < t < 2$ d. $0 < t < 2$

Q2: Prove mathematically (in detail) that the following signal is $x(t) = e^{j[(\frac{\pi}{2})t-1]}$ (8 Marks)

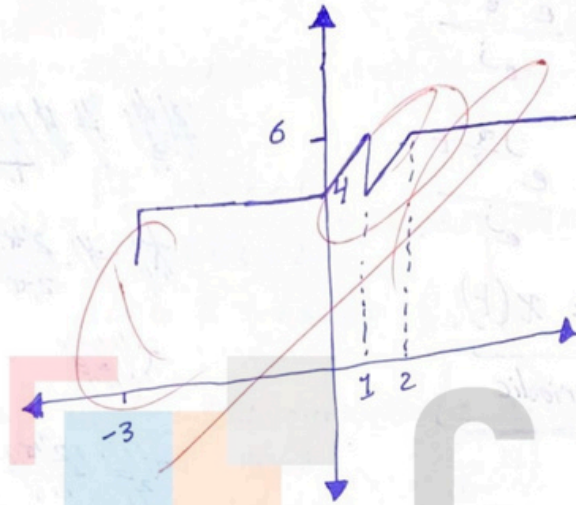
1. Periodic or aperiodic signal?
2. Energy or power signal?
3. Even or odd signal? neither
4. Causal or non-causal signal?

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Q3:

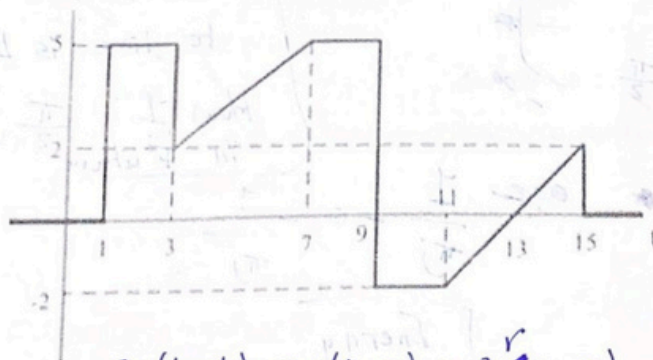
- I. Sketch $f(t)$ as detailed in the following equation: (4 Marks)

$$f(t) = u(t+3) + 2r(t) - u(t-1) - 2r(t-2) + 3$$



- II. write down the signal in mathematical form (3 Marks)

$x(t)$



$$x(t) = 5u(t-1) - 3u(t-3) + \frac{3}{4}r(t-3) - \frac{3}{4}r(t-7) - 7u(t-9) + r(t-11) - 2u(t-15)$$

$$\frac{3}{4}r(t-3) - \frac{3}{4}r(t-7)$$